

Context-Dependent Selection of Established Urban Routing Policies: Regime Effects, Decision Value, and Generalization Limits

Anonymous Author(s)

Anonymous institution(s)

Abstract. Navigation systems implement established routing objectives — shortest distance, reactive travel time, capacity-aware load balancing, reliability-aware routing — but commit to one by design rather than by traffic condition. We ask whether that commitment should be made per context, and separate two questions usually merged: whether the preferred policy varies with conditions, and whether exploiting that variation is worth anything.

In 12,960 microsimulation runs over 648 contexts (5 paired seeds per policy) spanning demand, signal timing, penetration, information lag, disruption and alternative capacity, the preference does vary: three of four policies are strictly best somewhere by a seed-resolved margin, and 73.0% of contexts have a non-singleton Pareto set over journey time, CO₂ and stopped delay — with the reliability-aware policy minimising stopped delay in 66% of contexts while rarely minimising journey time. Exploiting it is nevertheless worth 0.451 s per vehicle, 0.095% of the single-best fixed policy’s mean journey time, because that policy’s average gain where it wins is 18× its average loss where it does not; selecting the wrong fixed policy costs 46.4% of the same baseline.

Within that margin, a selector predicting the winner’s identity is *more accurate and more costly* than the fixed policy, whereas one predicting each policy’s *advantage* closes 55.8% of the available gap. Across 8 shift splits the unguarded selector helps on 2, is indistinguishable on 3 and harms on 3; a support-and-confidence gate recovered the fixed policy exactly on every harmful split, at the cost of the in-distribution gain, and one split shows a shift the gate cannot see because it leaves the feature space unchanged. The benchmark separates three questions usually treated as one: complementarity, decision value, and deployability. We introduce no routing, learning or multi-criteria decision algorithm.

Keywords: Routing-policy selection · Algorithm selection · Decision regret · Microsimulation · Distribution shift · Selective prediction.

1 Introduction

A guidance system makes two decisions, not one.

Level 1. Which routing *policy* should be active under the current traffic conditions?

Level 2. Given that policy, which route does each guided vehicle receive?

Level 2 is the subject of a large literature. This paper concerns Level 1, and treats the policies themselves as fixed, published objectives.

The question matters because established policies embody different assumptions. Distance minimisation assumes the shortest corridor can absorb what is sent to it. Reactive travel-time routing assumes recent measurements predict the near future, which fails when a large guided cohort acts on them simultaneously [2,1]. Load balancing assumes capacity is unevenly distributed and worth equalising. Reliability-aware routing assumes travel-time variability is itself a cost. Each assumption holds in some conditions and not others, so the policy that should be active plausibly depends on the traffic — and production systems nevertheless fix it at design time.

The gap this paper addresses. Showing that the preferred policy changes with conditions is not the same as showing that a context-aware selector is worth building. A winner map can vary substantially while the cost of ignoring it remains negligible, if the best fixed policy loses only slightly wherever it is not best. These are separate empirical quantities, and conflating them is easy because the first is the more visible. We therefore measure both, and a third: whether a selector that captures the available value can be trusted outside the conditions it was trained on.

Research questions. **RQ1** Under which traffic, signal, information, penetration, capacity and disruption conditions do established routing policies exhibit complementary strengths? **RQ2** Are the preference boundaries explicable by mechanistic traffic variables rather than only by a statistical model? **RQ3** Does a learned selector identify useful selection structure beyond a strong mechanistic rule? **RQ4** When the operating context lies outside the training support, or uncertainty is too large, can the system abstain safely?

Contributions.

1. A factorial characterisation of the operating regions in which four established routing policies are complementary, over 648 contexts and 12,960 paired microsimulation runs.
2. A mechanistic account of the preference boundary in dimensionless traffic descriptors, reported as brackets between sampled levels.
3. A decision-oriented evaluation in which the primary comparison is a learned selector against a mechanistic rule, not against a fixed policy, and in which selection accuracy and decision regret are shown to rank methods differently.
4. A selective mechanism with a support gate and a confidence gate, evaluated separately under interpolation, extrapolation, concept shift and domain shift, including a case it does not detect.

We introduce no routing algorithm, no learning algorithm and no multi-criteria decision method; every policy and model class used here is established

and cited. The contribution is the measurement and its interpretation. We make no priority claim: each component below has precedent, and the positioning in Section 2.4 states what is and is not new.

2 Related Work

2.1 Dynamic and Traffic-Aware Routing

Route choice under congestion rests on the equilibrium tradition founded by Wardrop [21]. Reactive guidance — routing on recently measured link travel times — is the operational descendant of that tradition, and its central difficulty is well documented: information supplied to drivers need not reduce congestion, because the response to the information changes the state it described [2]. Acemoglu et al. [1] give this a sharp game-theoretic form, showing that additional information can strictly worsen equilibrium outcomes. The severity depends on how much of the demand responds, which makes guidance penetration a first-order variable rather than an implementation detail [23].

Two policy families respond to these failures directly. Load-balancing guidance distributes vehicles over a set of travel-time-acceptable alternatives according to predicted link load; we use the flow-balanced k -shortest-path construction of Pan et al. [13]. Reliability-aware guidance adds a dispersion term to expected travel time, following the mean-variance formulation of Sen et al. [15]. Both are established families with published objectives, and we take them as given.

2.2 Environmental, Multi-Criteria, and Context-Dependent Routing

Eco-routing minimises an emissions proxy rather than time [4], and raises the question of whether an environmental objective is *distinct* from a temporal one in a given network. That is an empirical property of the environment, not a property of the objective, and we treat it as a measurement rather than an assumption (Section 5.10).

More generally, comparing routing policies on a single scalar hides the structure that multiple criteria expose. Where several criteria disagree, the identity of the preferred policy depends on which is chosen, and a Pareto analysis makes that dependence visible before any weighting is applied. We adopt that ordering — dominance first, scalarisation only as sensitivity — and avoid monetary valuation entirely, since no verified value of time, value of reliability or carbon price was available for this setting.

2.3 Algorithm Selection and Decision-Oriented Evaluation

Choosing among algorithms by instance features is Rice’s algorithm-selection problem [14], with a mature methodology developed largely in combinatorial search [22,18,9] and standard reference points — the virtual best and single best

solver — from benchmarking practice [3]. Instance-space analysis asks where in a feature space each algorithm is strong [17], which is precisely the form RQ1 takes for routing policies.

Two further strands matter here. First, evaluating a predictor by the cost of the decision it induces rather than by its predictive error is the predict-then-optimize distinction of Elmachet and Grigas [6]; our results show the two criteria ranking methods differently. Second, declining to decide is classical [5], and split conformal prediction supplies distribution-free intervals to decide when [20,11]. Its coverage guarantee rests on exchangeability, which covariate shift breaks [16,19] — a limitation we treat as a design constraint rather than a caveat.

2.4 Positioning

The literature supplies the routing objectives, the traffic-feedback mechanisms that make them fail, the algorithm-selection framework, the regret and virtual-best reference points, and the uncertainty tools — each separately and each well developed. This study combines them into a single empirical policy-selection evaluation, run under a controlled factorial of traffic conditions in a physical microsimulation, and reports decision regret rather than selection accuracy as the primary quantity.

We do not claim that regime-dependent routing has not been studied, nor that no prior work has combined any subset of these ideas. What we offer is the measurement: how much decision value the combination actually carries in a setting where every component is held fixed and known.

3 Problem Formulation

Objects. A *context* $x \in \mathcal{X}$ is a set of operating conditions known before a policy is activated: demand, signal timing, guidance penetration, information lag, disruption regime and alternative capacity. A *policy* $a \in \mathcal{A} = \{P1, \dots, P4\}$ maps individual vehicles to routes according to its own published objective. Running policy a in context x produces an outcome vector; we write $C(x, a)$ for the scalar decision cost, system mean journey time, and treat the other two criteria by dominance (Section 4.4).

The two levels. The decision studied here is the map

$$\pi : \mathcal{X} \rightarrow \mathcal{A}, \quad x \xrightarrow{\pi} a \xrightarrow{\text{policy } a} \text{routes} \longrightarrow \text{traffic outcome}. \quad (1)$$

Level 1 is π ; level 2 is the policy’s own route assignment, which is fixed and not learned. This is not route prediction, not vehicle-level action learning, not signal control, and not joint routing-signal optimisation.

Reference points. Two quantities from the algorithm-selection literature [3] bound what any π can achieve. The *single-best policy* (SBS) is the one fixed policy with the lowest mean cost, chosen on training contexts only:

$$a_{\text{SBS}} = \arg \min_{a \in \mathcal{A}} \mathbb{E}_{x \in \mathcal{X}_{\text{train}}} [C(x, a)]. \quad (2)$$

It is what a deployed fixed-policy system achieves, and the baseline any selector must beat to be worth building. The *virtual-best policy* (VBS) chooses per context using that context’s realised outcomes, $C_{\text{VBS}}(x) = \min_a C(x, a)$. *VBS is retrospective: it is not an algorithm, cannot be deployed, and is used only as an upper reference.*

Regret and headroom. The regret of selecting a in context x , and the mean regret of a selector, are

$$R(x, a) = C(x, a) - \min_{b \in \mathcal{A}} C(x, b), \quad \bar{R}(\pi) = \mathbb{E}_x [R(x, \pi(x))]. \quad (3)$$

The *headroom* is the total decision value available to any selector:

$$H = \bar{R}(a_{\text{SBS}}) = \mathbb{E}_x [C(x, a_{\text{SBS}}) - \min_b C(x, b)]. \quad (4)$$

H is the quantity that decides whether context-dependent selection is worth attempting at all, and it is measurable without building a selector. We report it in seconds per vehicle and, where a percentage is given, always as a share of $\mathbb{E}_x[C(x, a_{\text{SBS}})]$, the single-best policy’s mean journey time, stated alongside.

Three distinct questions. The paper keeps apart three properties that are often merged:

complementarity $\neq$ decision value $\neq$ deployability.

Complementarity asks whether $\arg \min_a C(x, a)$ varies with x . Decision value asks how large H is. Deployability asks whether an estimate $\hat{\pi}$ built from finite data retains that value outside the conditions it was fitted on. A portfolio can be strongly complementary, carry almost no decision value, and still produce a selector that is unsafe under shift — which is what we observe.

4 Methodology

4.1 Simulation Environment and Network

All experiments run in SUMO [12]; CO₂ comes from its HBEFA3 implementation [10]. Policies act through the simulator’s online control interface, so every routing decision is taken inside the running simulation from information available at that instant.

Figure 1 shows the environment: one origin–destination pair served by three structurally different corridors between a common diverge and merge, crossed

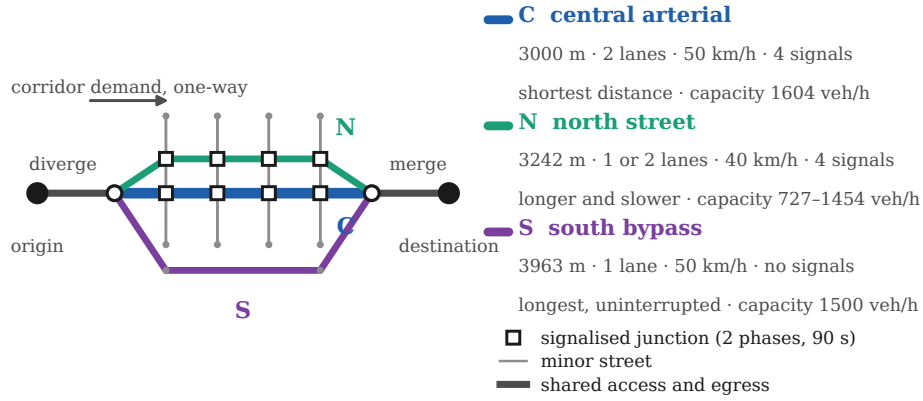


Fig. 1. The scenario network. One origin–destination pair is served by three structurally different corridors between a common diverge and merge, crossed by four minor streets carrying independent demand through the same signals. Geometry, lane counts and speed limits are read from the built network file; corridor capacities are measured rather than assumed.

by four minor streets carrying independent demand through the same signals, so a policy that loads a corridor also delays traffic it does not route. Because every junction permits straight-through movements only, the loopless path set is exactly these three — the candidate catalogue is complete, not a k -shortest approximation.

Two structural choices were fixed before the experiment, both identified by capacity validation. The destination portal gives each approach a dedicated exit lane, so an alternative corridor is not penalised by junction priority rather than by its own capacity; and the diverge carries no turn-radius speed ceiling, which in schematic node geometry would cap the bypass at a value belonging to the drawing rather than to the road.

Measured capacity. Each corridor was driven to saturation alone and its discharge measured: 1761 veh/h/lane on the arterial and 1596 veh/h/lane on the slower street, consistent to within 1.8% across green ratios, and 1500 veh/h on the uninterrupted bypass — below queue-discharge saturation flow, as free-flow headways require. The demand levels of Table 2 follow from these measurements rather than from a textbook constant.

4.2 Routing Policy Portfolio

Table 1 gives each policy’s objective, information requirement, intended strength and expected failure mode. *The failure modes were specified before the experiment*, and each factor exists to make one of them reachable: P1 → capacity shortage on the shortest corridor; P2 → information lag and herding; P3 →

Table 1. The routing-policy portfolio. Every policy is an established family; none is introduced here. The last column states a hypothesis fixed before the experiment.

	Objective and information	Intended strength	Hypothesised failure mode
P1	$\min_p L(p)$; no traffic state	no infrastructure, cannot oscillate	loads the shortest corridor past its capacity
P2	$\min_p \hat{\mu}_p(t - \Delta)$; lagged link travel times	follows congestion as it forms	moves the whole guided cohort on stale information
P3	$\min_{p \in \mathcal{A}} \max_{e \in p} \rho_e$; lagged times, flows and own assignments	spreads load when capacity is uneven	diverts to longer corridors with no capacity shortage
P4	$\min_p [\hat{\mu}_p + \lambda \hat{\sigma}_p]$; lagged link travel times	avoids corridors with volatile travel time	pays a detour for variance carrying no risk

$\mathcal{A}$: paths within a declared travel-time tolerance of the best. ρ_e : link utilisation. Δ : information lag. λ : dispersion weight.

unevenly distributed alternative capacity; P4 $\rightarrow$ travel-time variability under disruption.

All four read traffic state from one shared buffer: a policy acting at time t may read link travel times sampled no later than $t - \Delta$ over a 120 s window, falling back to free-flow values before any admissible sample exists. No policy sees the future or any realised outcome of the run it is part of. Policy parameters were frozen before the experiment ($\varepsilon = 0.20$ and a 120 s assignment window for P3, $\lambda = 1.0$ for P4).

Each policy was validated before use by twelve known-answer tests covering the observation lag, the free-flow fallback, and each policy’s defining behaviour. A two-path integration test inside the simulator, with equal path lengths and a 2:1 capacity ratio, confirms that P3 splits demand 1.97 against that ratio while P1 commits the entire cohort to one path.

4.3 Experimental Factors and Common Random Numbers

Six factors (Table 2) are crossed fully: 648 contexts, 5 seeds, four policies, 12,960 runs. The demand levels span uncongested to oversaturated conditions relative to the measured capacity (Figure 2).

The vehicle set, departure times, guided/unguided labelling, habitual corridor choice, background traffic and the incident realisation depend on the seed and context alone, never on the policy, so two runs of a context under different policies face identical vehicles and every comparison is paired. Unguided drivers follow a habitual corridor drawn once from a frozen logit over free-flow path times; at penetration p , a fraction $1 - p$ of corridor demand is therefore inert with respect to the policy under test.

Table 2. Scenario factors and levels. Demand levels are expressed as a fraction of the measured network capacity, which ranges from 0.31 to 1.10 at the median capacity configuration. Full factorial: 648 contexts.

Factor	Levels	Mechanism activated
Demand (veh/h)	1200, 1800, 2400, 3000, 3600, 4200	capacity shortage
Green ratio g/C	0.35, 0.50, 0.65	signalised capacity
Penetration	0.2, 0.5, 0.8	cohort size; herding
Information lag Δ (s)	30, 120, 300	staleness; oscillation
Incident regime	none, arterial	travel-time risk
Alt. capacity	1, 2 lanes	load to balance

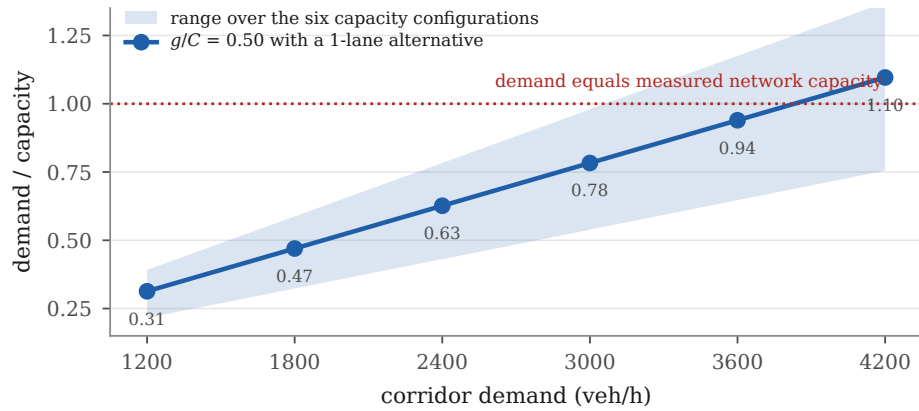


Fig. 2. Demand levels against measured network capacity. The shaded band spans the six capacity configurations produced by the green-ratio and alternative-capacity factors; the line marks the median configuration. The grid spans uncongested to over-saturated conditions, and the levels follow from measurement rather than from a chosen range.

Disruption as risk, not as a known event. When the incident regime is active, one lane closure is drawn — location, onset and duration — from a declared distribution seeded by the run seed. Within a context and seed the realisation is identical across policies, preserving the pairing; across seeds it differs, so the context carries travel-time risk no policy can know in advance. An incident occurring identically in every replication would be a known event rather than a risk, and could not test a reliability-aware policy.

Statistical treatment. The experimental unit is the context. An advantage of a over b is *resolved* only when the paired seed differences satisfy $|\bar{d}| > 2\text{SE}(\bar{d})$; unresolved differences are treated as ties, so no claim rests on a difference the seeds cannot separate.

Table 3. The three decision criteria. All are measured over the same cohort: every vehicle in the network, guided or not, whose intended departure falls in the measurement window.

Criterion	Measurement
C1 Mean journey time (s/veh)	mean over the cohort of insertion delay plus in-network time. Insertion delay is included: a policy that oversaturates the network produces vehicles that cannot enter it, and excluding their wait would credit the policy for the queue it caused.
C2 CO ₂ emissions (kg)	total CO ₂ over the cohort, HBEFA3 passenger-car petrol Euro-4 class.
C3 Total stopped delay (veh s)	total time spent below 0.1 m/s — the time vehicles spend <i>stationary</i> in queues, as distinct from the time they spend travelling slowly.

4.4 Outcome Criteria and Pareto Evaluation

Three criteria (Table 3) capture different burdens. **C1 mean journey time** is the travel burden per vehicle, including insertion delay — a policy that oversaturates the network produces vehicles that cannot enter it, and excluding their wait would credit the policy for the queue it caused. **C2 CO₂** is the environmental outcome. **C3 total stopped delay** is the stationary burden, time below 0.1 m/s in queues, distinct from time spent travelling slowly.

All three are measured over every vehicle in the network, guided or not, with intended departure in the measurement window, so a policy cannot be rewarded for helping guided vehicles at the expense of background traffic. Policies are compared by **Pareto dominance first**; the scalar cost used for regret is C1, which requires no weights. Scalarisation appears only as sensitivity, through three declared normalised preference profiles. No monetary value of time, value of reliability or carbon price is used: none was available from a verified source, and inventing one would make the conclusion an artefact of the invented number.

4.5 Mechanistic Rule

Each context is expressed through eight dimensionless descriptors computable before activation: network saturation x_1 ; shortest-corridor saturation $x_2 = D/\text{cap}_C$; alternative capacity share x_3 ; effective green ratio x_4 ; penetration x_5 ; information staleness $x_6 = \Delta/T_C^{\text{ff}}$; incident regime x_7 ; and controlled-flow saturation $x_8 = pD/\text{cap}_C$, the share of the shortest corridor’s capacity the guided cohort alone would consume. Capacities are the measured values, so x_1 , x_2 and x_8 are genuine degrees of saturation. A route-overlap index is omitted: the corridors share only access and egress edges, so overlap is structurally constant here.

The mechanistic rule B1 is an axis-aligned region rule on x_1 – x_8 of depth at most two, fitted on training contexts only. Thresholds are reported as *brackets*

between adjacent sampled levels, since a grid cannot locate a boundary more finely than its own spacing.

4.6 Context-Aware Selector

The learned selector predicts *advantage*, not identity:

$$\Delta_a(x) = C(x, a_{\text{SBS}}) - C(x, a), \quad \hat{\pi}(x) = \arg \max_a \hat{\Delta}_a(x), \quad (5)$$

with a_{SBS} determined on training contexts only. A positive Δ_a means a beats the fixed policy a deployed system would otherwise run, so the learned quantity is the one a decision-maker needs. The model is gradient-boosted decision trees [7,8], one regressor per policy, with hyperparameters fixed before evaluation and never tuned on a test fold. With a few hundred context-level instances and full feedback from every policy, higher capacity is neither needed nor defensible; we use no graph neural network, deep network, reinforcement learning or contextual bandit.

The ladder is B0 fixed policy, B1 mechanistic rule, B2 depth-3 tree, B3 multinomial logistic regression, B4 the advantage selector and B5 its selective variant, with VBS as reference. **The primary comparison is B4 against B1:** B0 is a low bar, and beating it shows only that the decision varies with context at all.

4.7 Validation, Uncertainty, and Distribution Shift

Evaluation is leave-one-context-out. Inside each fold, standardisation, the identity of the SBS, tree thresholds, conformal quantiles and the support envelope are recomputed from training data alone. No seed of a held-out context appears in training, and no realised outcome is used as a feature.

Two gates. B5 acts only if both pass. The *support gate* measures mean distance to the five nearest training contexts in standardised descriptor space and abstains above the 95th percentile of the training fold’s own distances; it is a feature-space extrapolation detector. The *confidence gate* is a split-conformal interval on the predicted advantage ($\alpha = 0.10$, calibrated on 30% of the training fold); the selector acts only when the lower bound on the advantage over the fixed policy is positive, and it is calibrated under exchangeability, hence in-distribution. If either gate fails, the fixed policy is used. Neither gate detects concept shift, and the conformal guarantee does not survive covariate shift [16,19], so we report gate behaviour rather than coverage guarantees outside the training distribution.

Shift splits. Eight held-out splits are evaluated separately and never pooled: demand above and below the training range, an interior demand level as an interpolation control, unseen information lag, unseen penetration, unseen green ratio, training without disruption and testing under it, and a disruption on a corridor that carries none in training. Pooling them would average an interpolation control with a domain shift and hide the only thing the experiment can say, which is that they differ.

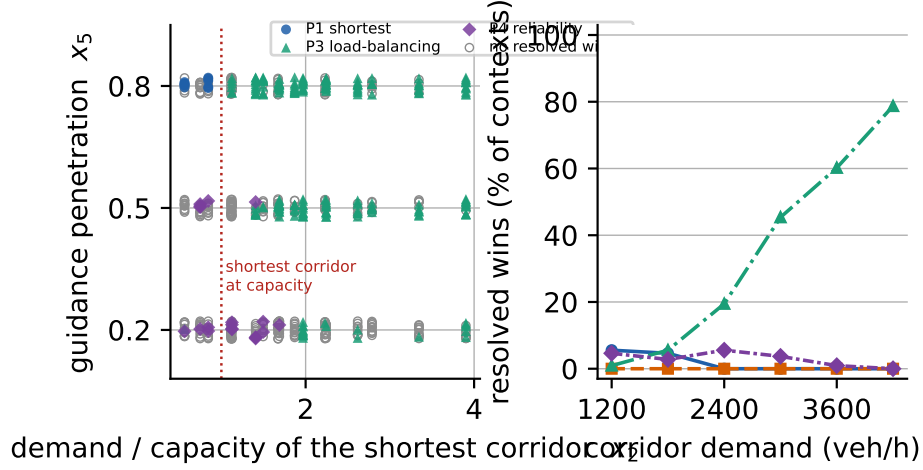


Fig. 3. The regime map. Left: the resolved winner in the plane of shortest-corridor saturation against guidance penetration; open circles mark contexts where the seeds resolve no winner. Right: the share of contexts each policy wins, by demand level.

5 Results

5.1 Campaign Validity

The main campaign executed 12,960 runs over 648 contexts with 4 policies and 5 paired seeds. No run failed. Every run completed every vehicle it generated (minimum completion 1.000), with 0 teleports in the entire campaign, so no result rests on a vehicle the simulator removed. All 648 contexts satisfy the validity and completeness rules declared in advance; none is excluded.

5.2 Policy Complementarity

The confirmatory criterion $G1'$ — declared before the campaign and reported here whichever way it came out — passes: 3 policies (P1, P3, P4) are strictly best in at least one context with a seed-resolved margin. **P2 is never a resolved winner anywhere in the factor space.** Resolved winner counts are P3 227, P4 19 and P1 11 (Table 4).

The winner map is multi-factor but shallow. Naming the same policy everywhere is correct in 64.0% of contexts; the best single factor (demand) reaches 67.0%, and the best pair 71.0%. The structure is interpretable and matches the design: P1 wins only where the shortest corridor is uncongested *and* penetration is high enough for adaptive policies to over-divert; P4 wins at low demand with generous green; P3 wins every context at the highest demand level (Figure 3).

Table 4. Policy complementarity over 648 contexts. A win is *resolved* when the paired seed difference against the runner-up exceeds twice its standard error. Pareto membership is over all three criteria.

	Best on C1		Non-dominated	
	any	resolved	n	share
P1	21	11	21	3.2%
P2	80	0	266	41.0%
P3	415	227	555	85.6%
P4	132	19	525	81.0%
<i>Pairwise C1 differences (s/veh); positive = first policy slower</i>				
Pair	mean	5-95% range	identical	
P1-P2	+203.87	[-0.3, +711.9]	0.0%	
P1-P3	+220.21	[+0.8, +761.6]	0.0%	
P1-P4	+201.87	[-1.0, +711.9]	0.0%	
P2-P3	+16.34	[-0.7, +75.9]	22.7%	
P2-P4	-2.00	[-18.2, +9.6]	42.6%	
P3-P4	-18.34	[-87.3, +1.3]	20.2%	

5.3 Decision Value of Adaptation

Complementarity and decision value are different quantities, and here they point in opposite directions.

The single-best fixed policy is P3, optimal in 65% of contexts. Figure 4 shows why: the distance-minimising policy diverges with demand while the three adaptive policies track one another. Running the single-best policy always costs 474.4 s of system mean journey time per vehicle; the retrospective best-per-context choice costs 473.9 s. The headroom of Eq. 4 is therefore

$$H = 0.451 \text{ s per vehicle}, \quad \frac{H}{\mathbb{E}_x[C(x, a_{\text{SBS}})]} = \frac{0.451}{474.4} = 0.095\%. \quad (6)$$

Selecting the wrong *fixed* policy costs far more. Using the distance-minimising policy throughout increases system mean journey time by 220.2 s per vehicle relative to the single-best fixed policy, which is 46.4% of that policy’s mean journey time of 474.4 s. The reactive and reliability-aware policies cost +16.3 s (+3.4%) and +18.3 s (+3.9%) on the same basis.

Choosing the fixed policy well is worth 46.4% of mean journey time. Choosing it *per context*, given that the best fixed policy is already in use, is worth 0.095%.

5.4 Why the Adaptation Value Is Small

The explanation is not that the policies behave alike; Section 5.2 shows they do not. It is that the best fixed policy is near-dominant in a measurable sense.

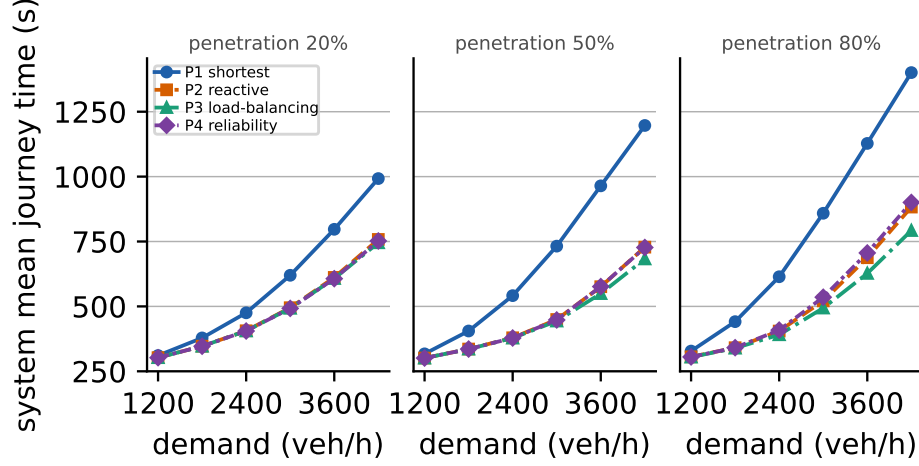


Fig. 4. System mean journey time against demand at each guidance penetration level, averaged over the remaining factors. The distance-minimising policy diverges once the shortest corridor saturates, while the three adaptive policies track one another closely — the visual form of the small decision value quantified in Section 5.4.

Define the *downside* of the fixed policy as its mean loss in those contexts where it is not best, and its *upside* as its mean gain over the best alternative in those contexts where it is. In 34.3% of contexts P3 is not best, and there it loses 1.32 s on average (at most 17.2 s). In 64.0% of contexts it is best, and there it wins by 23.1 s on average (at most 153.5 s). Its upside is **18× its downside**.

A portfolio containing such a member has a large ranking structure and a small decision structure: any selector moving away from it risks a large loss to chase a small gain. This is an empirical characterisation of this portfolio in this environment, not a general theorem — but it is measurable in any portfolio, before a selector is built.

5.5 Noise and Resolution of the Adaptation Effect

Among the 222 contexts where the fixed policy is not best, the mean headroom is 1.32 s against a mean two-standard-error band of 1.67 s on the same paired comparison, and the headroom exceeds its own noise in 55 of them (25%). Resolving the campaign-wide mean effect of 0.451 s would require roughly 25 seeds per context, about 64,800 runs. We report this rather than presenting a near-noise-floor improvement as an achievement.

5.6 Mechanistic Rule

Fitted on all contexts for exposition, the depth-2 rule splits first on $x_8 = pD/\text{cap}_C$, the share of the shortest corridor’s capacity the guided cohort alone

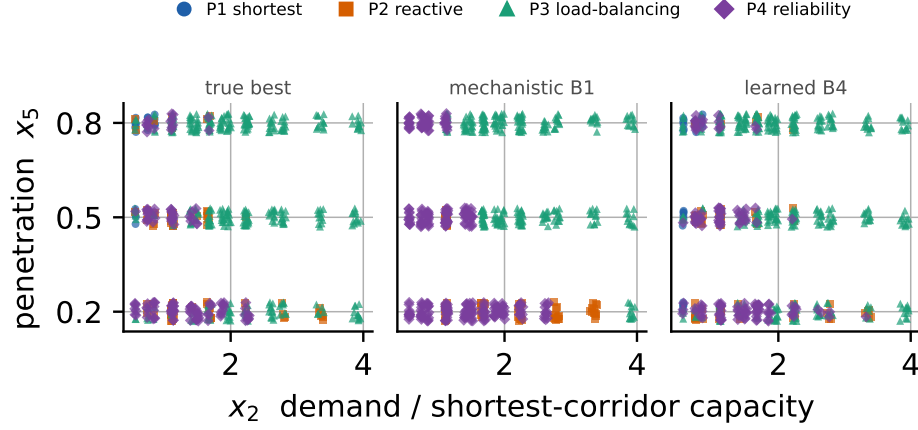


Fig. 5. Selection regions in the mechanistic feature space: the true best policy, the depth-2 mechanistic rule, and the advantage-based selector, over controlled-flow saturation and guidance penetration.

would consume, at a threshold bracketed between 0.748 and 0.781. The rule is physically readable: below that share the decision turns on the effective green ratio; above it, on whether the arterial is already oversaturated (Figure 5).

The rule identifies the best policy in 68.8% of contexts, more often than always naming the fixed policy. Its mean decision regret is nevertheless 0.668 s against 0.451 s for the fixed policy. **It is more accurate and more costly.** Under leave-one-context-out the same pattern holds: B1 reaches 67% top-1 accuracy against B0’s 65%, and 0.77 s of regret against 0.45 s.

This is the predict-then-optimize distinction [6] in a traffic setting. A rule trained to name the winner is rewarded for being right where the margin is negligible and punished only mildly for being wrong where it is large — the wrong trade under an $18\times$ asymmetry.

5.7 Advantage-Based Selector

Predicting advantage (Eq. 5) rather than identity reverses the failure. B4 reaches 0.20 s of mean regret, closing 55.8% of the gap between the fixed policy and the retrospective best, while being *less* accurate at naming the winner (73%) than the logistic baseline (76%, 0.28 s). It also has the lowest maximum regret of any selector, 7.0 s against 17.2 s (Table 5, Figure 6).

The primary comparison declared in advance was B4 against B1, and B4 wins it decisively: 0.20 s against 0.77 s. Beating B0 alone would not have been evidence of much, since B0 is near-dominant by Section 5.4.

The magnitude is what matters operationally. B4 reduces mean regret by 0.252 s per vehicle relative to the fixed policy — 0.053% of a 474.4 s mean

Table 5. Decision quality under leave-one-context-out evaluation. Regret is in seconds of system mean journey time per vehicle. “Within noise” is the share of contexts whose regret falls below that context’s own seed noise floor.

		Regret (s/veh)				in	gap
		mean	med.	max	CVaR ₉₀	noise	closed
B0	best fixed policy	0.45	0.00	17.2	3.7	99%	—
B1	mechanistic rule	0.77	0.00	17.2	6.5	96%	-0.714
B2	tree, depth 3	0.34	0.00	17.2	3.0	99%	0.238
B3	logistic	0.28	0.00	17.2	2.5	100%	0.373
B4	GBDT advantage	0.20	0.00	7.0	1.7	100%	0.558
B5	selective, abstaining	0.45	0.00	17.2	3.7	99%	0.000
	VBS (not deployable)	0.00	0.00	0.0	0.0	100%	1.000

journey time. The selector is real and reproducible, and below the threshold at which an operator would act on it.

5.8 Selective Prediction and Distribution Shift

B5 abstains in 36% of in-distribution contexts and returns exactly the fixed policy’s regret, 0.45 s. The mechanism is explicit: the conformal interval on the predicted advantage is wider than the advantage, so the requirement that its lower bound exceed zero is never met. A calibrated gate refuses to act on an effect smaller than its own uncertainty.

Across the 8 shift splits (Table 6, Figure 7), the unguarded selector B4 does not behave uniformly. Using a tolerance of 0.05 s per vehicle, it **helps** on 2 splits, is **indistinguishable** on 3, and **harms** on 3:

- *helps*: the interpolation control O3 (0.56 to 0.17 s) and, notably, the domain shift O8 (0.31 to 0.18 s);
- *indistinguishable*: unseen low demand O2, unseen lag O4, unseen penetration O5;
- *harms*: unseen high demand O1 (0.00 to 0.98 s), unseen green ratio O6, and worst of all the concept shift O7 (0.63 to 2.22 s).

On every one of the 3 harmful splits, B5 recovered the fixed policy’s regret exactly. The selective rule therefore prevented the observed harms; it does not guarantee protection against harms not observed here.

The support gate flags 100%, 100% and 100% of held-out contexts as outside the training envelope on the lag, penetration and green-ratio extrapolations respectively. B1 degrades worst of all under shift, reaching 6.15 s on unseen penetration.

A shift the gate cannot see. O8 is instructive precisely because the support gate does *not* fire: it flags 0% of held-out contexts, because those contexts’ descriptors are identical to descriptors seen in training — the incident’s *location* is not among

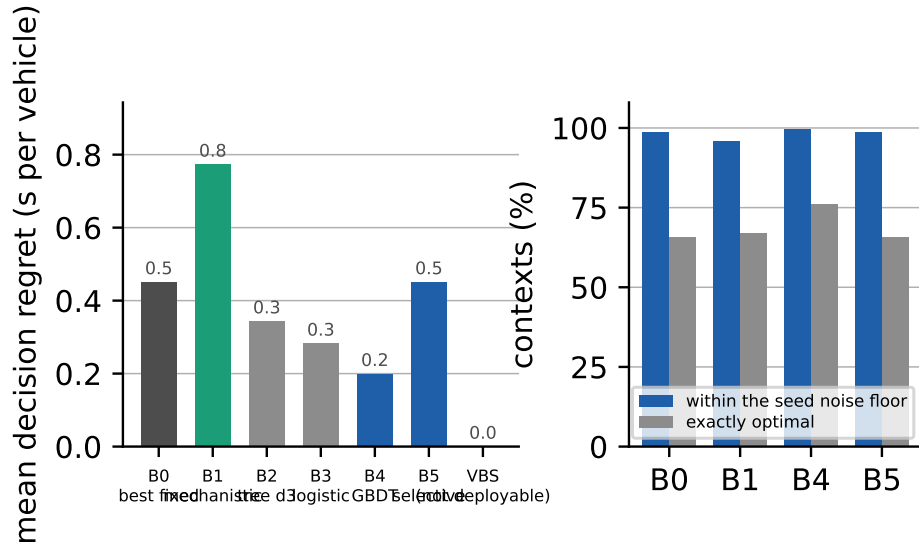


Fig. 6. Decision quality under leave-one-context-out evaluation. Left: mean regret for each selector; the retrospective best is hatched because it is not deployable. Right: the share of contexts in which each selector is within the seed noise floor, and exactly optimal.

the eight. The selector happens to do well there, but it does so unguarded. A support gate defined on a feature space cannot detect a shift that leaves that feature space unchanged, and this is a concrete instance rather than a theoretical caveat.

5.9 Boundary Refinement

Where the main grid showed the preferred policy changing between adjacent demand levels, three intermediate levels were run in each of the 19 widest-margin cells.

13 of 19 refined sequences are monotone, with a single change point, and in point estimate they narrow the switching bracket from the 600 veh/h grid spacing to a median of 150 veh/h. All 6 shortest-to-load-balancing transitions are monotone; their change points sit in a 150 veh/h bracket whose position differs by cell, from 1800–1950 veh/h to 2250–2400 veh/h. That the position moves with the cell is the point: the boundary is a property of the operating conditions, not a single number for the network.

None of these refined steps is seed-resolved. The point estimates move consistently, but no individual step between adjacent refined levels exceeds twice its own standard error at 5 seeds. The remaining 6 sequences are non-monotone, and all involve the reliability-aware policy. We therefore report the transition as a

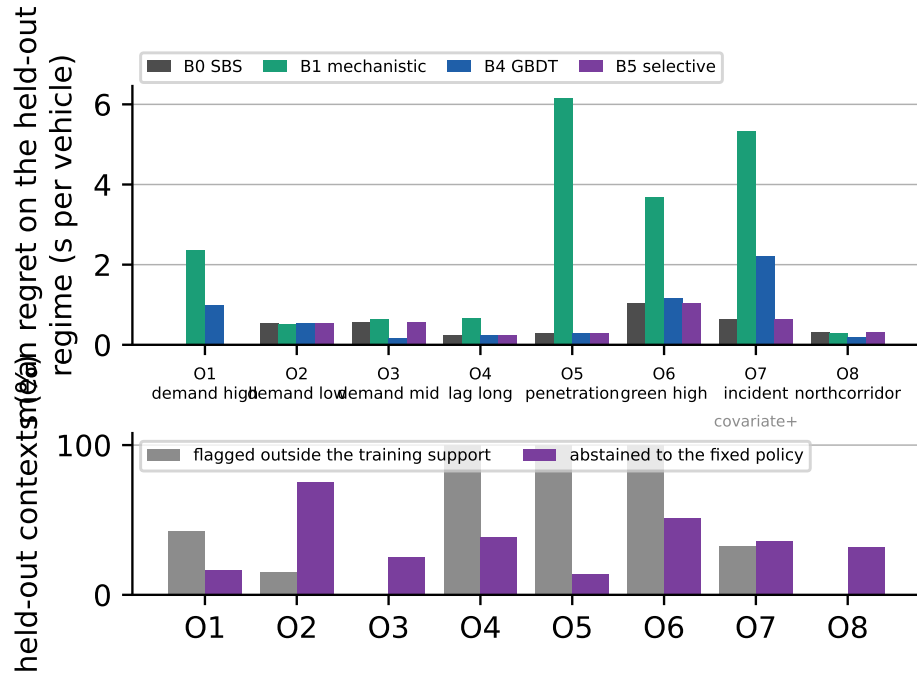


Fig. 7. Distribution shift, reported separately by shift type. Top: mean regret on the held-out regime for each selector. Bottom: the share of held-out contexts flagged outside the training support, and the share on which the selective selector abstained.

Table 6. Distribution shift, reported separately by shift type. Regret in seconds of system mean journey time per vehicle on the held-out regime. The support column is the share of held-out contexts the support gate flagged as outside the training envelope.

Held out	Shift type	B0	B1	B4	B5	flagged	abstained
O1 demand 4200	extrapolation	0.00	2.35	0.98	0.00	43%	17%
O2 demand 1200	extrapolation	0.53	0.52	0.53	0.53	15%	75%
O3 demand 2400	interpolation	0.56	0.64	0.17	0.56	0%	25%
O4 lag 300	extrapolation	0.23	0.66	0.24	0.23	100%	38%
O5 penetr. 0.8	extrapolation	0.30	6.15	0.29	0.30	100%	14%
O6 gc 0.65	extrapolation	1.04	3.68	1.16	1.04	100%	51%
O7 incident 1	cov.+concept	0.63	5.33	2.22	0.63	32%	36%
O8 incident on north corr.	domain shift	0.31	0.29	0.18	0.31	0%	31%

point-estimate bracket and state that it is unresolved: the shortest-path to load-balancing reversal is an ordered regime change, while the apparent reliability-aware transitions near it are not distinguishable from noise.

5.10 Three-Criterion Pareto Structure

Over all 648 contexts, 73.0% have a non-singleton Pareto set. Non-domination is not concentrated in one policy: P3 is non-dominated in 86% of contexts, P4 in 81%, P2 in 41% and P1 in 3%.

The criteria are not interchangeable. Journey time and CO₂ correlate at +0.9390 and rank the four policies identically in 52% of contexts; journey time and stopped delay correlate at +0.8916 and agree in only 34%. The best policy on CO₂ differs from the best on journey time in 30% of contexts; on stopped delay it differs in 56%.

The policies specialise by criterion rather than by regime. P3 minimises journey time in 418 contexts and P3 minimises CO₂ in 453, but **P4 minimises stopped delay in 429 contexts (66%)** while minimising journey time in only 129. The reliability-aware policy routes towards the steady, unsignalised corridor: its vehicles spend less time stationary and more time moving. **This is where the portfolio’s complementarity lives** — on the criterion vector, not on the scalar cost.

Eco-routing remains excluded, on evidence. CO₂ and journey time correlate at +0.9390 and select the same policy in 52% of contexts. An objective nearly monotone in travel time adds a label rather than a policy; the numbers are given so a reader may disagree with the threshold rather than with an assertion.

5.11 Sensitivity

Across the three declared preference profiles the preferred policy changes in 14.5% of contexts (Table 7). The journey-time choice is reproduced by the time-weighted profile in 93% of contexts, the balanced profile in 86% and the environment-weighted profile in 80%. The decision is stable under reasonable preference assumptions, and where it is not, stopped delay is what moves it.

Table 7. Sensitivity and robustness. Upper: the two policy parameters frozen before the experiment, the effect of changing each on that policy’s own mean journey time, and how often the identity of the best policy is unchanged. Lower: the three declared preference profiles over normalised criteria, the policy each selects, and how often that agrees with the journey-time choice.

<i>Policy parameters</i>						
Policy	Param.	Frozen	Alt.	Δ (s)	Δ (%)	winner same
P3	p3_eps	0.2	0.1	+8.10	+1.27%	84%
P3	p3_eps	0.2	0.3	-16.66	-2.62%	78%
P4	p4_lambda	1.0	0.5	-0.11	+0.00%	91%
P4	p4_lambda	1.0	1.5	+1.27	+0.16%	98%
<i>Preference profiles</i> (weights on normalised C1–C3)						
Profile	Weights	P1	P2	P3	P4	agrees C1
Time	(0.70, 0.15, 0.15)	19	76	406	147	92.9%
Balanced	(0.40, 0.30, 0.30)	8	70	399	171	86.0%
Environment	(0.20, 0.55, 0.25)	7	66	425	150	80.1%

The two frozen policy parameters behave differently. The dispersion weight of the reliability-aware policy is close to immaterial, changing that policy’s own mean journey time by at most 0.16%; across all four parameter changes the identity of the best policy is unchanged in 78% to 98% of the contexts tested. The admissibility tolerance of the load-balancing policy is not immaterial: loosening it from 0.20 to 0.30 improves that policy’s own mean journey time by 2.62% (16.7 s). The frozen value therefore *understates* how dominant the load-balancing policy is, which strengthens rather than weakens the central conclusion. We report this rather than retrofitting the better value.

6 Experimental Integrity and Deviations

What was frozen, and when. The experiment specification — factors and levels, criteria, seed count, the resolution criterion, screening gates, baseline ladder, model class and hyperparameters, abstention rules and shift splits — was written and committed to version control *before any evaluation run was executed*. Policy parameters were frozen at the same time and varied afterwards only as sensitivity (Section 5.11). No factor level was chosen because it produced a particular outcome; the demand levels follow from measured capacity.

Execution reconciliation. The main campaign comprises 12,960 valid runs ($648 \times 4 \times 5$), and this is the number underlying every result in Section 5 unless stated otherwise. The project executed 20,284 simulations in total, of which 18,124 succeeded: 384 scenario-validation, 640 screening, 12,960 main, 1,920 sensitivity, 1,080 domain-shift and 1,140 boundary-refinement runs, plus 2,160 runs that

failed for the reason given below. These counts are distinct and are not interchangeable.

Protocol deviation D-1. The pre-specified complementarity screen did not satisfy criterion G1 and therefore triggered the declared stopping rule. The main campaign was nevertheless executed, under the documented protocol deviation D-1. The reasons for continuation were recorded before campaign launch: a selection problem demonstrably existed; G1 was written against the scalar criterion and was blind to the three-criterion structure beside it; and the negative finding rested on 16 of 648 contexts sampled at extreme factor levels only. A confirmatory criterion G1', declared before the main campaign and identical in definition to G1, was then evaluated on all 648 contexts. G1' passes (Section 5.2).

We do not present G1' as erasing the deviation, and we do not describe the original screen as having passed. It did not. The methodological lesson is that the screen sampled corner conditions and did not sample the interior transition region, so it had power to detect near-ties but not coverage to find where the minority policies win. A resolution-IV fraction at extreme levels is an efficient design for main effects and a poor one for discovering where a winner map changes. That is a limitation of our screening design, not a justification for having continued.

An infeasible shift split and its replacement. One declared shift split placed the disruption on the bypass corridor. The bypass is single-lane, so the specified capacity-reducing lane closure does not constrain it — it disconnects it, leaving vehicles already routed there without a valid path. All 2,160 attempted runs failed for this reason. The split was replaced by the nearest feasible construction, a disruption on the two-lane north corridor, which preserves the intent (a corridor carrying no disruption in any training context) while remaining a capacity reduction rather than a severance. A guard now refuses any incident on a single-lane edge. The replacement is reported as split O8 and is *not* the originally declared split.

Traceability. Every numeric claim in this paper is generated from a single machine-readable results file by a script, and rendered through a macro; a build-time check fails if the text references a quantity the analysis did not produce. Seventeen automated quality-control checks cover leakage discipline, training-fold-only baseline selection, freeze ordering, shift reporting and citation completeness. All 23 references were verified against publisher records; DOIs that could not be confirmed directly were omitted rather than inferred.

7 Discussion

7.1 Complementarity Does Not Imply Adaptation Value

The portfolio is complementary: three policies win somewhere by a resolved margin, the winner map needs more than one factor, and 73.0% of contexts

have a non-singleton Pareto set. On the usual reading, that is the evidence for building a selector.

It is not sufficient evidence. The decision value of the entire winner map is 0.451 s per vehicle, 0.095% of the mean journey time under the single-best fixed policy, because that policy’s upside where it wins is $18\times$ its downside where it does not. The two quantities are computable from the same data, one is small when the other is large, and only the second determines whether adaptation is worth attempting.

The practical implication is a cheap first step. Before choosing a model class, compute the gap between the best fixed policy and the retrospective best, and compare it with the noise floor of that same paired comparison. Neither requires a model. Had we done so here, the answer would have been a 0.095% gap sitting close to the resolution of a 5-seed design.

7.2 Decision-Oriented Evaluation Changes the Interpretation of Accuracy

Selection accuracy and decision regret rank the methods differently in this study, and the disagreement is not marginal. The mechanistic rule is more accurate than the fixed policy (67% versus 65%) and more costly (0.77 s versus 0.45 s). The advantage-based selector is less accurate than the logistic baseline and cheaper than it.

The cause is not model capacity but the training target. Methods trained to name the winner optimise a loss that treats every context alike; a context where the margin is 20 s contributes no more than one where it is 0.2 s. Predicting advantage weights each context by what is actually at stake, which under an $18\times$ asymmetry is the difference between a selector that helps and one that hurts. This is the predict-then-optimise argument [6], appearing here not as a refinement but as a sign change.

For anyone reporting such a system: report decision regret, and do not present selection accuracy as a proxy for it.

7.3 A Strong Fixed Policy Can Be a Rational Default

In this benchmark the load-balancing policy is optimal in 65% of contexts and loses only 1.32 s on average where it is not. Against that, the distance-minimising policy costs 220.2 s per vehicle more — 46.4% of the single-best policy’s 474.4 s mean journey time.

Two decisions therefore sit three orders of magnitude apart: which fixed policy to run, and whether to vary it by context. A benchmark that reports only the second makes the first invisible.

7.4 Selective Adaptation and the Limits of Feature-Space Support

The selective rule prevented the observed harms: on each of the 3 harmful shift splits it recovered the fixed policy’s regret exactly. It did so by declining to

act — the conformal interval is wider than the effect — and the price is the in-distribution gain the unguarded selector achieves.

Its limits are equally concrete. The conformal gate is calibrated under exchangeability and so is an in-distribution instrument; the support gate is a feature-space extrapolation detector; neither detects concept shift. Split O8 shows what that means in practice: a disruption moved to a corridor carrying none in training left every descriptor unchanged, the support gate flagged 0% of those contexts, and the selector acted unguarded. It happened to do well. A support gate cannot detect a shift its feature space does not represent, and reporting the case is more useful than asserting the principle.

7.5 Implications for Routing-Policy Deployment

We state what this benchmark demonstrates, not what an operator should do.

The benchmark shows that in this environment the decision carrying the substantial value is the choice of fixed policy, and that the incremental value of context-dependent selection is small relative to it and close to the resolution of the experiment. It shows that if such a selector is built, the training target determines whether it helps, and that a support-and-confidence gate removed the harms observed under shift while also removing the gains. And it shows that the criterion matters as much as the policy: the policy minimising stopped delay is not the policy minimising journey time in 56% of contexts, which is a different argmin rather than a preference weighting.

Whether these carry to a real network is not established by anything here.

8 Limitations

Environment and scope. The network is synthetic: a controlled instrument built to isolate mechanisms, not a model of any city. Corridor demand is a single origin–destination pair choosing among three corridors; the four minor streets couple the routing decision to traffic that is not routed, but many simultaneously-routed pairs, with the interactions between their choices, are outside what is tested. No claim of real-world validation is made anywhere in this paper. What is portable is the method rather than the numbers: the descriptors are dimensionless, so a boundary is stated as a degree of saturation and a controlled-flow share, both measurable on a real network.

Decision context and compliance. The selector is given the operating conditions; a deployed system would have to forecast them and would inherit the forecast’s error on top of the selector’s. Guided vehicles comply fully with the policy, and unguided vehicles never respond to conditions at all; real behaviour lies between these. The reported decision quality is therefore an upper bound on what the same selector would achieve behind a forecast and partial compliance.

Resolution and statistical reach. The experiment samples a finite grid with 5 seeds. Boundaries are reported as brackets between adjacent sampled levels and never as continuous thresholds, and no refined switching step is seed-resolved (Section 5.9); resolving the campaign-wide adaptation effect would need roughly 25 seeds per context. Where a factor is associated with a change of preferred policy, that is an association within a designed factorial, not an identified causal mechanism.

Protocol. Two deviations are on the record and are described in Section 6: the screening gate G1 failed and the campaign proceeded under deviation D-1, and one declared shift split was infeasible and was replaced by the nearest feasible construction. Neither is presented as neutral.

Uncertainty machinery. Split conformal coverage requires exchangeability, which covariate shift breaks; the conformal gate is calibrated in-distribution and heuristic outside it, the support gate detects only extrapolation visible in the feature space, and neither detects concept shift. Split O8 is a demonstrated instance of a shift the support gate does not see.

Frozen parameters and the emission model. The policy parameters were fixed before the experiment and are not optima; a better-tuned load-balancing tolerance improves that policy by 2.62%, which strengthens the central conclusion but means the reported value is not the best available. CO₂ is computed by the simulator’s HBEFA3 implementation for one vehicle class, without fleet composition, cold starts or gradient, so C2 should be read as a consistent comparative index rather than an absolute emissions estimate. Finally, no monetary valuation is used anywhere, which keeps the study free of invented constants but also means it cannot express a trade-off in currency.

9 Conclusion

We characterised context-dependent selection among four established urban routing policies in 12,960 microsimulation runs over 648 controlled contexts, with every policy decision taken from information available before activation and every comparison paired under common random numbers.

The portfolio is complementary: three of four policies are strictly best somewhere by a seed-resolved margin, the winner map depends on more than one factor, and 73.0% of contexts have a non-singleton Pareto set over journey time, CO₂ and stopped delay. The reliability-aware policy, which rarely minimises journey time, minimises stopped delay in 66% of contexts. Complementarity in this portfolio lives in the criterion vector rather than in the scalar cost.

Exploiting it is nevertheless worth 0.451 s of system mean journey time per vehicle, 0.095% of the single-best fixed policy’s 474.4 s, because that policy’s upside where it wins is 18× its downside where it does not. Selecting the wrong fixed policy costs 220.2 s per vehicle, 46.4% of the same baseline.

Within that margin the methodological findings are clear. A selector trained to predict the winner’s identity is more accurate than the fixed policy and more costly than it; one trained to predict each policy’s advantage is less accurate and closes 55.8% of the available gap. Across 8 shift splits the ungarded selector helped on 2, was indistinguishable on 3 and harmed on 3, and a support-and-confidence gate recovered the fixed policy exactly on every harmful split while forgoing the in-distribution gain. One split showed a shift the support gate cannot see, because it left the feature space unchanged.

The benchmark’s contribution is to separate three questions usually treated as one: whether policies are complementary, whether adaptation carries decision value, and whether a selector can be deployed safely. Here the answers are yes, barely, and only with a gate that mostly declines to act. Both of the first two quantities are measurable before any model is built, and we suggest measuring them first.

Reproducibility

All scenario, policy, simulation and analysis code, every run record, and the scripts generating every number, table and figure are provided with the submission. The experiment specification was committed before any evaluation run was executed; both protocol deviations are documented in Section 6.

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